HCIA-AI V4.0 — Onsite Masterclass Core Concepts & Exam Traps Breakdown

Huawei ICT Academy Onsite Masterclass Notes • Core Concepts & Exam Traps • Curated by: Umed Jamal Nouri (@UmedJaMala)

PART 1: CORE EXAM QUESTIONS & TRAPS HIGHLIGHTED BY CERTIFIED INSTRUCTOR

1.1 The llama.cpp Real Exam Question

- **Instructor Exam Alert:** *'Do not underestimate anything on the slides! Questions specifically test standalone C/C++ runtimes like llama.cpp on official exams.'*
- **Technical Fact Tested: llama.cpp** is the ONLY C/C++ implementation tool mentioned in the HCIA-AI V4.0 curriculum.
- **Exact Exam Angle:** It allows developers to run Large Language Model (LLM) inference locally (on-premise) or in the cloud across diverse hardware architectures **WITHOUT needing an AI Development Framework** (such as PyTorch, TensorFlow, or MindSpore). Be careful if a True/False or Single-Choice question claims llama.cpp requires PyTorch/MindSpore to run!

1.2 The 'Feature Engineering (Optional)' True/False Exam Trap

- **Slide Observation:** In Stage A (Data Preparation), Step 3 is written as **Feature Engineering (Optional)**.
- **Why is it Optional?** Because in modern Deep Learning, **Computer Vision (CV)** and **Natural Language Processing (NLP)** models automatically extract hierarchical features via neural layers—they do NOT need manual Feature Engineering. However, **Recommendation Systems** still strictly require manual Feature Engineering to extract user behavior/item attributes.
- **Key Exam Trap Highlighted:** *'If the exam asks: We use Feature Engineering to collect important features in CV and NLP applications — the answer is FALSE! It is included in the pipeline specifically for Recommendation Systems.'*

1.3 The Two Different 'Evaluations' Trap (Pre-Evaluation vs. Post-Evaluation)

- **Key Exam Warning:** *'There are TWO evaluations in the AI business process—do not mix them up in the exam!'*
- **1st Evaluation — Pre-Evaluation (Data Evaluation, BEFORE Training):** Evaluates the **Dataset** itself to prevent Bias and Overfitting. It checks three pillars: **(1) Quality**, **(2) Diversity**, and **(3) Balance / Fairness**.
- **2nd Evaluation — Post-Evaluation (Model Evaluation, AFTER Training):** Evaluates the **Trained Model** on the Test Set ('Final Exam') using the 4 Classification Metrics (**Accuracy, Precision, Recall, F1-Score**) or 3 Regression Metrics (**MSE, RMSE, MAE**).

1.4 Regression Metrics & The Outlier Exam Question (MAE vs. MSE & RMSE)

- **MSE (Mean Squared Error):** Squares the difference between actual and predicted values: sum((y - y_hat)^2)/n. Because it squares the error, the unit scale is distorted (an error of 2 becomes 4).
- **RMSE (Root Mean Squared Error):** Takes the square root of MSE. **Exam Keyword:** RMSE is **more intuitive** than MSE because its output scale matches the original target unit (an error of 2 stays 2).
- **MAE (Mean Absolute Error):** Takes the absolute value |y - y_hat| without squaring.
- **CRITICAL EXAM QUESTION ON OUTLIERS:** What is an Outlier? An extreme anomalous data point. Because MSE and RMSE square the error, a single large Outlier explodes the squared error and distorts MSE/RMSE. Therefore, **MAE is the most robust and suitable metric when Outliers exist!** Instructors emphasize: *'Be careful, questions frequently test metric sensitivity to outliers!'*

1.5 Edge vs. Core Server Quantization (INT8 vs. FP16 vs. FP32)

- **Business Rule:** More floating-point bits = more computing power = **higher hardware cost**.

- **FP32 to FP16 (Half-Precision):** Reduces 32-bit floats to 16-bit floats; cuts memory usage in half while preserving high model accuracy.
- **FP32/FP16 to INT8 (Integer Quantization):** Completely eliminates the decimal/fractional part. Greatly improves power efficiency and inference speed.
- **Exam Distinction (Where to use INT8 vs. FP32): INT8 is ideal for Edge Devices** (e.g., a Smart Bicycle chipset that only checks whether the device is active and tracks basic runtime), and is **NOT ideal for Core/Cloud Servers**. Conversely, the central billing/core server uses **FP32** because financial and core calculations require full decimal precision.

1.6 Model Conversion to Middleware (.pt/.pth vs. .onnx)

- AI runs on 4 distinct processor architectures: **CPU, GPU, TPU, and NPU**.
- When you train a model in PyTorch, it saves a framework-specific file (**.pt / .pth**) tied to the training environment.
- To deploy across heterogeneous hardware/cloud platforms, you must perform **Model Conversion to Middleware** using the universal **.onnx (Open Neural Network Exchange)** file format.

1.7 High-Memorization Chapter Warning (CBK Datasets & Algorithms)

- Instructors emphasize that the **AI Business Process & Application Scenarios / Datasets** chapter is one of the most challenging parts of the certification exam because it contains specific dataset names, algorithm names, and acronyms that cannot be guessed by logic alone—**they must be memorized verbatim**.

PART 2: COMPLETE CHRONOLOGICAL LECTURE BREAKDOWN

Section 1 | Distributed Computing, Business Pipeline & Recommendation Systems

- **Distributed Computing (18 Chinese Universities Example):** No single university in China could afford a massive AI supercomputer alone, so 18 universities interconnected their data centers to share compute workloads.
- **llama.cpp (C++ Tool):** The sole C++ inference engine in the curriculum; runs LLMs locally or on the cloud without requiring an AI framework.
- **Business Perspective of AI:** Enterprise AI balances three constraints: **Cost, Time, and Quality**. Relies on Foundation/Large Models + Big Data + Fine-Tuning + **Prompt Engineering**.
- **The 4th Application Paradigm & 6/7 Business Steps:** After Desktop, Web, and Mobile apps, AI Applications represent the 4th paradigm. Step 1 (Requirement Analysis) must answer two core questions: *What is the purpose of the application?* and *What is the problem to be solved?*
- **Domain 1 — Recommendation Systems:**
- Predicts user demand from browsing history, purchases, ratings, and HTTP Cookies.
- **Key Limitation of Recommendation Systems:** User **Privacy** concerns due to continuous behavioral tracking.
- **4 Recommendation Algorithm Types (MUST MEMORIZE):** (1) **Collaborative Filtering**, (2) **Content-Based Recommendation**, (3) **Knowledge-Based Recommendation**, (4) **Hybrid Recommendation**.
- **3 Main Scenarios:** E-commerce product recommendation (Shein/AliExpress), Personalized News push (Google News), Video/Music recommendation (YouTube).
- **Domain 2 — Computer Vision (CV):** The 'Artificial Eye'—enabling machines to interpret visual data, prominently used in security and public safety surveillance.

Section 2 | CV Datasets, Medical AI, Autonomous Driving & 6G Sensing

- **Facial & Target Recognition:**
- **3 Scenarios:** (1) Identity Authentication, (2) Social Media Filtering, (3) Security Management.
- **Datasets (MEMORIZE): LFW (Labeled Faces in the Wild)** and **MS-Celeb-1M**.
- **Algorithms (MEMORIZE): CNN, FaceNet, ArcFace**.
- **Image Classification & Medical Diagnosis:**

- Addresses clinical misdiagnosis through automated radiological and pathological screening.
- **Datasets (MEMORIZE): ImageNet**, **COCO (Common Objects in Context)**, and **ChestX-ray14** (medical chest X-rays).
- **Algorithms (MEMORIZE): ResNet, VGG, Inception series**.

• **Autonomous Driving (L3/L4 vs. L5), 5.5G/6G & LiDAR:**

- Commercial autonomous vehicles today operate at **L3 or L4**; full **L5** autonomy has not been achieved yet.
- **3 Core CV Tasks:** Road Sign Recognition, Pedestrian Detection, and Obstacle Avoidance.
- **Datasets (MEMORIZE): KITTI, Cityscapes, Waymo**.
- **Algorithms:** 2D Visual Detection: **YOLO, SSD, Faster R-CNN**. 3D Point-Cloud Sensing: **PointNet, RangeNet++**.
- **5G vs. 6G Exam Concept: 5G is Cloud-Native**, whereas **6G will be AI-Native** (built on **Sensing** and **Digital Twin Networks**). Explains **LiDAR** (3D laser pulse distance mapping) and **Wi-Fi Sensing**.

Section 3 | NLP Applications, Voice Security & Stage A (Data Preparation)

• **Chatbots, Smart Speakers & Voice Security:**

- Customer Service Assistants and Smart Speakers automate enterprise support workflows.
- Highlights security risks of AI **Voice Synthesis / Cloning** and social engineering reconnaissance.
- **Dialogue Algorithms (Sequential Models): Seq2Seq, Transformer, BERT, GPT, Reinforcement Learning**.

• **Sentiment Analysis:**

- Extracts emotional polarity and user attitudes from text or facial expressions.
- **Datasets (MEMORIZE): IMDB Movie Review Dataset**, **Twitter Sentiment Analysis Dataset**.
- **Models (MEMORIZE):** Traditional ML: **Bag of Words (BoW), TF-IDF**. Deep Learning: **TextCNN, LSTM, Bi-LSTM**.

• **Machine Translation & Speech Processing (STT / TTS):**

- **Dataset: WMT (Workshop on Machine Translation)**. **Models: Transformer, XLNet, BERT**.
- Speech encompasses **Speech-to-Text (STT)** and **Text-to-Speech (TTS)**.

• **Stage A — Data Preparation (7 Sequential Sub-steps):**

1. **Data Collection:** Recommendation (purchases, browsing, ratings), CV (ImageNet, COCO, Pascal VOC), NLP (web, books, social media).
2. **Data Cleansing:** Removing duplicate records, irrelevant samples, typos, and missing values.
3. **Feature Engineering (Optional):** Required ONLY for Recommendation Systems (NOT needed for CV/NLP).
4. **Data Labeling:** Assigning ground-truth annotations for Supervised Learning.
5. **Data Splitting:** Dividing into **Training Set** (model learning), **Validation Set** (hyperparameter tuning & error reduction during training), and **Test Set** (final generalization evaluation).
6. **Preprocessing (Normalization & Encoding): Normalization** scales continuous numerical values into a standard range (e.g., [0,1]); **Encoding** converts categorical strings (e.g., Male/Female) into numerical values (0/1) to optimize memory and compute cost.

Section 4 | Data Evaluation, Model Hierarchy, 6 Hyperparameters & Backpropagation

• **Sub-step 7 — Data Analysis & Evaluation (Pre-Evaluation):**

- Checks dataset **Quality, Diversity, and Balance/Fairness** prior to training to prevent algorithmic bias.

• **Stage B — Model Selection & Hierarchy Design:**

- Model selection depends on **Task Type** and **Data Characteristics** (CV: VGG/ResNet/YOLO; NLP: RNN/LSTM/GRU/BERT/GPT).
- **Attention Mechanism:** Concentrates computational weight on the most relevant input features for the target objective.
- **4 Layer Types in Model Hierarchy:** (1) **Input Layer**, (2) **Hidden Layers**, (3) **Output Layer**, (4) **Auxiliary Layers (Pooling & Normalization)**.

• **The 6 Core Hyperparameters (Set Manually BEFORE Training):**

1. **Learning Rate:** Controls the step size / speed of parameter updates.
2. **Batch Size:** Number of samples processed per iteration within one training **Epoch**.
3. **Regularization Strength:** Penalizes complexity to overcome **Overfitting**.
4. **Number of Layers:** Network depth.
5. **Number of Nodes:** Neurons per hidden layer.
6. **Dropout Rate:** Randomly deactivates redundant nodes during training to improve **Generalization**.

• **Stage C — Model Training, Chain Rule & Gradient Issues:**

- **Forward vs. Backward Propagation:** Forward pass computes predictions and the **Loss Function**; Backward pass (Backpropagation) updates **Weights (W)** using partial derivatives (dZ/dW) via the calculus **Chain Rule**.
- **Gradient Explosion vs. Gradient Vanishing:** High-order power derivatives multiply into massive values causing **Gradient Explosion**, whereas repeated derivatives of small fractions decay toward zero before reaching early layers, causing **Gradient Vanishing**.

Section 5 | Post-Evaluation Metrics, Model Deployment & Exam-Day Strategy

• **Stage D — Model Evaluation (Classification & Regression):**

- **Classification Metrics: Accuracy** (overall correctness), **Precision** = TP/(TP+FP) (positive prediction quality), **Recall** = TP/(TP+FN) (sensitivity to actual positives), and **F1-Score** (harmonic mean of Precision and Recall).
- **Regression Metrics: MSE**, **RMSE** (intuitive unit scale), and **MAE** (most robust to Outliers).

• **Stage E — Model Deployment (3 Core Steps):**

1. **Model Conversion to Middleware:** Exporting `.pt/.pth` into universal **`.onnx`** format for CPU/GPU/TPU/NPU deployment.
2. **Model Quantization:** Compressing `FP32 -> FP16` or `FP32 -> INT8` (Edge devices use `INT8`; Core servers use `FP32`).
3. **Model Encapsulation:** Packaging the model behind **REST APIs / Sockets** for system integration.

• **Official Exam-Day Strategy (60 Questions in 90 Minutes):**

- With 60 questions in 90 minutes (~1.5 minutes per question) and long 3–4 line English stems, never stall on a difficult item. Select a tentative answer, click the **Bookmark / Flag** button, and use the final 15 minutes on the summary screen to review flagged items.
- Exercise strict caution on **Multiple-Answer Questions** where incorrect selections penalize the score.

Section 6 | Foundation Models, Emergence & Prompt Engineering (From Latest Onsite Lecture)

• **Philosophy of Machine Learning & Generalization:**

- Machine learning is not rote memorization. The fundamental objective of parameter optimization is **Generalization**: developing an internal representation that reliably distinguishes correct from incorrect outputs on *unseen* test data.

• **The Emergence Phenomenon in Foundation Models (Micro vs. Macro):**

- **Emergence Definition:** When a complex neural architecture scales in parameter count and dataset volume, the microscopic interactions between billions of individual weights produce qualitative **Macro-level breakthrough capabilities** that cannot be anticipated by analyzing single components in isolation.
- **The River Flood Analogy (From Lecture):** Micro teams inspect individual stream branches and rainfall levels, while the Macro team manages the unified emergent outcome (the flood). In LLMs, billions of micro-weights produce macro-level emergent intelligence.
- **Crucial Exam Trap:** Emergence is non-linear. Emergent abilities do not scale smoothly from small models; they appear abruptly once a critical threshold of compute, data, and parameters is crossed.

• **Prompting Paradigms: In-Context Learning (ICL) vs. Chain-of-Thought (CoT):**

- **In-Context Learning (ICL):** Few-shot prompting where input-output demonstrations are provided directly inside the prompt context window without parameter updates. Excellent for text classification, translation, and formatting, but fails when confronted with multi-step reasoning.

- **Chain-of-Thought (CoT):** Prompts the model to generate intermediate reasoning steps ('Let us think step by step'). CoT explicitly teaches the model *the sequence of logic and intermediate deductions* required to reach the correct answer in mathematical, algorithmic, and symbolic reasoning tasks.

• **The 4 Strategic Pillars of Modern Foundation Models (DeepSeek & Huawei Pangu):**

1. **Open Source:** Shatters closed proprietary monopolies and accelerates global developer collaboration.
2. **Cost-Effective:** Maximizes compute efficiency so training and inference costs are reduced by orders of magnitude.
3. **High Performance:** Algorithmic architectural innovations match or exceed closed frontier models.
4. **Lower Development Barrier:** Directly overcomes global chip and compute shortages through resource-efficient architectures.

• **Huawei AI Application Ecosystem & Celia (Xiaoyi - □□):**

- The ultimate objective of AI is business deployment. Huawei integrates its **Pangu Foundation Models** into industrial pipelines and consumer hardware via **Huawei Celia (Xiaoyi)**, powering multimodal intelligent interactions across HarmonyOS and Ascend compute infrastructure.